A Task-First Audit of Dynamic LiDAR World Simulation

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Abstract

Dynamic LiDAR world simulation needs explicit build-only inputs, held-out targets, visibility semantics, and baselines that separate surface quality from the return operator. We audit two routes: reusable object-surface completion and full neural LiDAR simulation. We introduce a versioned ray-preserving data contract, recover 3,882 keyframes for dependency-chain-isolated development data, and evaluate raw fusion, Actor-local TSDF, a legacy learned surface, and AdaPoinTr under common density caps and ray metrics. An official LiDAR4D run establishes complete range, return, intensity, and scan capability on KITTI-360. On a clean four-log development set with 501 Actors, our pre-registered observation-constrained AdaPoinTr candidate underperforms Actor-local TSDF in Chamfer distance, F-score, early returns, and hit recall. We therefore close the method claim before route selection or source-test access. The result is a reproducible negative audit that identifies strong simple baselines and prevents a narrower, unsupported method claim.

1. Introduction

Actor insertion pipelines can produce plausible point sets while violating the sensing process that generated them. Evaluation becomes especially misleading when held-out returns influence the forward pass, output density differs across methods, or a learned readout hides weak geometry. A useful simulator must distinguish invalid and censored queries, valid no-returns, background occlusion, and Actor returns while composing all surfaces in world-depth order.

We organize the study around that task rather than around an inherited model. Our first route asks whether mature point completion gains physical consistency from measured hit and free-space constraints. Our second asks whether the broader problem instead requires a complete neural LiDAR model. Both face deterministic baselines and external methods before any final test is opened.

This audit contributes (i) a target-free, ray-preserving Actor data contract with log-level role controls; (ii) a matched geometry and conditional-return evaluation that exposes density and readout effects; (iii) executed AdaPoinTr and LiDAR4D baselines; and (iv) a pre-registered negative route decision on 501 clean-development Actors. The negative result is central: the proposed observation-constrained completion does not beat Actor-local TSDF, so we stop before route selection, source testing, and application claims.

2. Related Work

Point-cloud completion. Object-level completion predicts a denser or more complete surface from partial points; PoinTr provides a strong transformer baseline for this setting [2]. It is a relevant baseline family only when inference remains independent of held-out targets and all methods share the same return operator.

Occupancy completion. Occupancy methods predict volumetric free/occupied structure. They are not directly equivalent to a point-surface representation and require volume supervision before an occupancy claim is justified.

Neural LiDAR simulation. Neural field and ray-tracing systems such as LiDAR4D [1] jointly model geometry, intensity, ray drop, and visibility. They provide a broader route, but typically require scene-specific training, tracked dynamic objects, compiled CUDA operators, and substantially more compute.

3. Task Definition

Let the build evidence for an actor be E, including observed points, sensor origins, frame provenance, and known free/occupied evidence. For a query ray $q = (o, d, r_{\min}, r_{\max})$, a predictor may use only (E, q) at inference. Held-out return depths and points are labels and never enter the forward path.

The output state belongs to {invalid, censored, no-return, background-occluded, actor-return, other}. When an actor return is predicted, the model also reports a world depth and optional intensity. Scene composition selects the nearest valid actor or background surface by world depth. A valid no-return does not, by itself, certify the entire ray as free space.

4. Methods Under Audit

4.1. Evidence Bundle

The versioned Actor bundle stores build points, ray origins and directions, timestamps, sensor and ring identifiers, intensity validity, Actor and sensor poses, and local evidence. Query targets are stored separately. Code-level role gates prevent source and external tests from being opened before selection.

4.2. Shared Surface-to-Return Operator

Each object method outputs a surface. A common operator measures bidirectional point distance and intersects held-out rays with that surface using fixed lateral and depth tolerances. Geometry and conditional-return metrics remain separate, and every density cap reports the actual point count when a native surface contains fewer points.

4.3. Object-Surface Candidate

AdaPoinTr is first adapted from its official PCN checkpoint using the legacy training partition. The candidate is then fine-tuned for 100 epochs with its surface loss plus a hit-depth Huber term and a known-free early-surface penalty, both computed from the same predicted point set. We fix both added weights to 0.1 without a sweep. Before reading clean development quality, the primary output is fixed to 75% measured TSDF points and 25% learned completion points; the unanchored network is diagnostic.

4.4. Full-LiDAR Capability

The broader route uses the official LiDAR4D implementation on KITTI-360, including its scene representation, range and intensity rendering, ray-drop refinement, and scan export. This run establishes end-to-end capability. Because its task and denominator differ from object completion, it cannot serve as a numerical route-selection shortcut.

5. Experiments

Protocol. Historically exposed Actors are used only for D0 diagnostics and training. Four dependency-chain-isolated nuScenes logs form development; three further logs are frozen for route selection and three remain unopened source-test candidates. We selectively extract 3,882 official keyframe scans and materialize 501 development Actors with 641,930 held-out rays. All surface predictions are target-free. The D1 rule, fixed before development quality is read, requires either a 5% relative CD reduction or a 2-point F-score gain over the strongest baseline, an early-return increase no larger than one point, and a hit-recall loss no larger than one point.

D0 baselines. On the exposed diagnostic cohort, raw fusion and Actor-local TSDF form a non-dominated simple frontier. AdaPoinTr benefits substantially from official pretraining relative to training from scratch, but remains behind that frontier at every fixed 64–512 point budget. A scalar return-weight head recovers nearly all benefit of the capacity-matched F/O/U head, while F/O/U auxiliary supervision does not improve consistently across geometries. These results close the broad three-state representation claim and motivate only the narrower observation-constraint test.

Table 1. Clean development results under the 512-point cap. Native surfaces with fewer points are not duplicated; their mean realized size is reported.

Method	Pts.	CD↓	F↑	Early↓	Hit↑
Raw fusion	266	.1857	.7439	.4091	.5237
Actor TSDF	401	**.1786**	**.7503**	.4115	**.5497**
AdaPoinTr	512	.2292	.6344	.4332	.4910
Obs.-constrained	512	.2298	.6326	.4290	.4929
Anchored candidate	449	.1950	.7145	.4393	.5222

D1 result. Table 1 reports the clean gate. Relative to Actor TSDF, the anchored candidate changes CD by $+9.17\%$, F-score by -3.58 points, early returns by $+2.79$ points, and hit recall by -2.76 points. Thus neither primary-effect path nor either side-effect guard passes. The unanchored candidate reduces early returns by only 0.43 points and raises hit recall by 0.18 points relative to AdaPoinTr while slightly reducing geometric quality.

Full-LiDAR boundary. The official LiDAR4D run on KITTI-360 frames 4950–5000 completes 30k nominal training iterations, 1000 ray-drop refinement steps, evaluation, and scan export. It obtains ray-drop F1 .9553, depth RMSE 2.9472 m, and point CD/F-score .1173/.9208 in 8791.6 seconds on one RTX 3090. This supports complete-LiDAR capability but supplies no matched method gain for D1. The frozen decision is therefore negative D1 closure; route-selection and final roles remain unread.

6. Limitations

Only four fully isolated source logs are available for development, so uncertainty across driving domains is not resolved. We intentionally do not read the three route-selection logs or source-test candidates after the development failure. The object task contains positive held-out returns and measured pre-hit free samples, but it does not support a full volumetric occupancy claim. LiDAR4D is evaluated on its native KITTI-360 protocol; its absolute metrics are not comparable with the object table. The study therefore validates the audit, data, and baseline infrastructure while providing no competitive method, application, generalization, occupancy, or safety claim.

7. Conclusion

A task-first protocol changes the conclusion of this project. Strong deterministic baselines, density controls, and isolated development data show that the proposed observation-constrained completion does not improve the object surface or its physical return behavior. A full neural LiDAR method is executable but has no matched incremental candidate here. We therefore close the current method claim and preserve the remaining test roles. Future work should begin with a new hypothesis and new independent evidence, rather than tune this failed gate.

References

[1] Shengyuan Huang, Zan Gojcic, Mikhail Usvyatsov, Andreas Wieser, and Konrad Schindler. Lidar4d: Dynamic neural fields for novel space-time view lidar synthesis, 2023. 1

[2] Xumin Yu, Yongming Rao, Ziyi Wang, Zuyan Liu, Jiwen Lu, and Jie Zhou. Pointr: Diverse point cloud completion with geometry-aware transformers, 2021. 1